



COURSE SLIDES

Microsoft Certified Azure AI Engineer Associate AI-102 Training

TGS Ref No: TGS-2023036651

TSC: Artificial Intelligence Application in Product Development (ICT-TEM-4034-1.1)

Conducted by Tertiary Infotech Academy Pte Ltd | UEN 201200696W

Version 1.0 | 13 Jul 2026

COURSE ADMINISTRATION

Welcome and Housekeeping

Digital Attendance (Mandatory)

Take AM, PM and Assessment digital attendance for WSQ-funded courses.

Scan the SSG portal QR code shown by the trainer or administrator.

Minimum 75% attendance is required for funding and assessment eligibility.

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on AI agents, automation (n8n) and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

About the Trainer



Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

Let's Know Each Other

Your name and organisation or role.

Your experience with Azure or AI services.

Your goal for this AI-102 course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively - no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

75% attendance is required.

Lesson Plan - 2 Days

Day 1

Planning, security, monitoring, responsible AI.
Generative AI, Foundry, Azure OpenAI, RAG.
Agentic AI and Foundry Agent Service.
Vision, video, NLP, speech, translation.

Day 2

Custom language and question answering.
Azure AI Search, knowledge mining, vector search.
Document Intelligence and Content Understanding.
Capstone review and final assessment.

Skills Framework

TSC Title: Artificial Intelligence Application in Product Development.

TSC Code: ICT-TEM-4034-1.1.

Abilities: analyse algorithms, evaluate AI applications, assess feasibility and improvements.

Knowledge: AI applications, performance analysis, evaluation methods, algorithm design and implementation.

Learning Outcomes

Analyse Azure AI applications and establish their correlation with efficiency.

Identify and evaluate strengths and limitations of Microsoft Azure AI applications.

Assess feasibility and improvements when applying Azure AI to product and maintenance processes.

Course Outline

- Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution
- Lab 02: Responsible AI, Content Safety, Governance
- Lab 03: Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG
- Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts
- Lab 05: Computer Vision, Custom Vision, Video Insights
- Lab 06: NLP, Language, Speech, Translation
- Lab 07: Custom Language Models and Question Answering
- Lab 08: Azure AI Search, Knowledge Mining, Vector Search
- Lab 09: Document Intelligence and Content Understanding
- Lab 10: AI-102 Capstone and Course Review

Assessment

Written Assessment - Short-Answer Questions (WA-SAQ).

Practical Performance (PP).

Open book assessment: slides, Learner Guide and approved materials only.

Final assessment starts on Day 2 at 4:00 PM.

Briefing for Assessment

Place phones and other materials under the table or on the floor.

No photos or recording of assessment scripts.

No discussion during assessment.

Submit assessment scripts when time is up.

Criteria for Funding

Minimum attendance rate of 75% based on SSG Digital Attendance record.
Complete the assessment and be assessed as Competent.

LMS / TMS and Labs

LMS/TMS: <https://lms-tms.tertiaryinfotech.com>

Course labs: GitHub labs folder for TGS-2023036651.

Use the lab sequence in this deck, Learner Guide and Lesson Plan.

AI-102 FOUNDATIONS

Day 1

Planning, responsible AI, generative AI, agents, vision and language.

Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution

Main topic: Planning and operations

Scenario: A company wants a customer support AI platform with document search, answer generation, sentiment analysis, speech transcription, and image processing. You must plan the Azure AI resources.

Validation: You should have a service map, security plan, monitoring plan, and architecture diagram.

Lab 01: Objectives

Select appropriate Azure AI services.

Plan resource deployment and endpoints.

Identify authentication, keys, monitoring, and cost controls.

Prepare an AI solution architecture.

Lab 01: Imported Reference Diagram



Azure OpenAI

The Azure OpenAI service provides access to OpenAI generative AI models including the GPT family of large and small language models and DALL-E image-generation models within a scalable and securable cloud service on Azure.



Azure AI Vision

The Azure AI Vision service provides a set of models and APIs that you can use to implement common computer vision functionality in an application. With the AI Vision service, you can detect common objects in images, generate captions, descriptions, and tags based on image contents, and read text in images.



Azure AI Speech

The Azure AI Speech service provides APIs that you can use to implement *text to speech* and *speech to text* transformation, as well as specialized speech-based capabilities like speaker recognition and translation.



Azure AI Language

The Azure AI Language service provides models and APIs that you can use to analyze natural language text and perform tasks such as entity extraction, sentiment analysis, and summarization. The AI Language service also provides functionality to help you build conversational language models and question answering solutions.

How to use it

Connect the diagram to the lab scenario: A company wants a customer support AI platform with document search, answer generation, sentiment analysis, sp...

Use it as a reference while completing the design steps.

Imported from source PPT slide 25.

FOLLOW IN ORDER

Lab 01: Step Sequence

1. Map Requirements to Services
2. Plan Foundry Resources
3. Plan Security
4. Plan Monitoring and Cost
5. Draw Architecture

Lab 01 Step 1: Map Requirements to Services

Plan, Manage, Monitor, and Secure an Azure AI Solution

Requirement	Service
Generate responses	Azure OpenAI in Foundry Models
Search internal documents	Azure AI Search
Analyze sentiment	Azure AI Language
Transcribe calls	Azure AI Speech
Extract text from images	Azure AI Vision
Extract invoice fields	Azure AI Document Intelligence

Record the required output in your lab notes and verify it before continuing.

Lab 01 Step 2: Plan Foundry Resources

Plan, Manage, Monitor, and Secure an Azure AI Solution

Document:

Hub or project name
Region
Model deployment
Default endpoint
Connected data source
Responsible AI settings

Record the required output in your lab notes and verify it before continuing.

Lab 01 Step 3: Plan Security

Plan, Manage, Monitor, and Secure an Azure AI Solution

Record:

- Role-based access control.
- Managed identities.
- Key protection.
- Private endpoint or network controls.
- Logging and diagnostics.
- Secrets storage.

Record the required output in your lab notes and verify it before continuing.

Lab 01 Step 4: Plan Monitoring and Cost

Plan, Manage, Monitor, and Secure an Azure AI Solution

Document:

Token or transaction usage
Latency
Errors
Content safety events
Budget alerts
Diagnostic logs

Record the required output in your lab notes and verify it before continuing.

Lab 01 Step 5: Draw Architecture

Plan, Manage, Monitor, and Secure an Azure AI Solution

Use diagrams.net:

Application -> API layer -> Foundry/Azure AI services -> Search index/storage -> Monitoring

Record the required output in your lab notes and verify it before continuing.

Lab 01: Validation and Checkpoints

You should have a service map, security plan, monitoring plan, and architecture diagram.

Why is service selection part of AI engineering?

Why should keys be protected?

What should be monitored in an AI service?

What is the role of responsible AI planning?

Lab 02: Responsible AI, Content Safety, Governance

Main topic: Responsible AI

Scenario: The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.

Validation: You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

Lab 02: Objectives

Apply responsible AI principles.

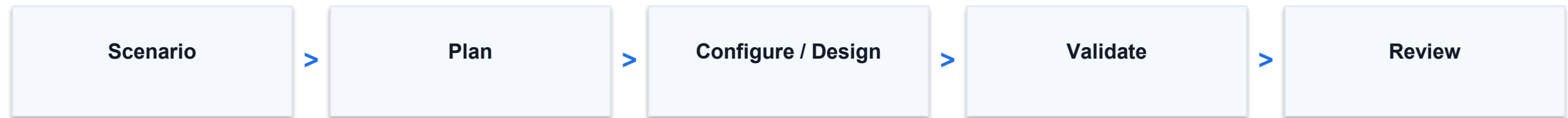
Configure content safety concepts.

Explain content filters, blocklists, prompt shields, and harm detection.

Design a responsible AI governance framework.

Lab 02: Lab Alignment Diagram

Lab-aligned design flow



Scenario

The support AI assistant may receive abusive user input, produce inaccurate answers, or expose sensitive data. You must design responsible AI controls.

Lab steps

- Review Responsible AI Principles
- Define Content Safety Controls
- Create a Safety Policy
- Test Governance Scenarios
- Create an Incident Plan

Validation

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

Lab 02: Step Sequence

1. Review Responsible AI Principles
2. Define Content Safety Controls
3. Create a Safety Policy
4. Test Governance Scenarios
5. Create an Incident Plan

Lab 02 Step 1: Review Responsible AI Principles

Responsible AI, Content Safety, Governance

Write one control for:

Fairness
Reliability and safety
Privacy and security
Inclusiveness
Transparency
Accountability

Record the required output in your lab notes and verify it before continuing.

Lab 02 Step 2: Define Content Safety Controls

Responsible AI, Content Safety, Governance

Document:

- Harm categories.
- Severity levels.
- Content filters.
- Blocklists.
- Prompt shields.
- Groundedness checks.
- Human escalation rules.

Record the required output in your lab notes and verify it before continuing.

Lab 02 Step 3: Create a Safety Policy

Responsible AI, Content Safety, Governance

Write:

Allowed use:
Disallowed use:
Blocked content:
Human review trigger:
Audit log requirement:
Escalation owner:
Review frequency:

Record the required output in your lab notes and verify it before continuing.

Lab 02 Step 4: Test Governance Scenarios

Responsible AI, Content Safety, Governance

Decide what should happen when:

- User asks for harmful instructions.
- Model produces low-confidence answer.
- Prompt contains personal data.
- Retrieved document is outdated.
- User asks for legal advice.

Record the required output in your lab notes and verify it before continuing.

Lab 02 Step 5: Create an Incident Plan

Responsible AI, Content Safety, Governance

Define how to report, investigate, remediate, and prevent repeat AI safety incidents.

Record the required output in your lab notes and verify it before continuing.

Lab 02: Validation and Checkpoints

You should have responsible AI controls, safety policy, scenario decisions, and incident plan.

What are content filters?

What is a prompt shield?

Why is human escalation important?

Who is accountable for AI output?

Lab 03: Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Main topic: Generative AI

Scenario: The company wants a grounded assistant that answers questions from internal support articles and generates draft replies.

Validation: You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.

Lab 03: Objectives

Plan a generative AI solution with Microsoft Foundry.

Select and deploy a model conceptually.

Design prompt templates and prompt flow.

Implement a RAG pattern conceptually.

Evaluate model and flow outputs.

Lab 03: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
The company wants a grounded assistant that answers questions from internal support articles and generates dra...

Use it as a reference while completing the design steps.

Imported from source PPT slide 309.

Lab 03: Step Sequence

1. Select a Model
2. Design Prompt Templates
3. Plan Prompt Flow
4. Design RAG
5. Evaluate Outputs

Lab 03 Step 1: Select a Model

Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Record:

Model name
Use case fit
Context window
Latency
Cost
Safety settings
Deployment option

Record the required output in your lab notes and verify it before continuing.

Lab 03 Step 2: Design Prompt Templates

Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Create templates for:

- Direct answer.
- Answer with sources.
- Escalation when context is missing.

Record the required output in your lab notes and verify it before continuing.

Lab 03 Step 3: Plan Prompt Flow

Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Draw:

User question -> classify intent -> retrieve context -> prompt template -> model -> safety check -> response

Record the required output in your lab notes and verify it before continuing.

Lab 03 Step 4: Design RAG

Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Document:

- Source documents.
- Cleaning.
- Chunking.
- Embeddings.
- Vector store.
- Retrieval filters.
- Citation format.
- Evaluation method.

Record the required output in your lab notes and verify it before continuing.

Lab 03 Step 5: Evaluate Outputs

Generative AI, Foundry, Azure OpenAI, Prompt Flow, RAG

Score:

Criterion	Score 1-5
Correctness	
Grounding	
Helpfulness	
Safety	
Conciseness	

Record the required output in your lab notes and verify it before continuing.

Lab 03: Validation and Checkpoints

You should have model selection notes, prompt templates, prompt flow diagram, RAG design, and evaluation table.

What is RAG?

Why are prompt templates useful?

What does tracing help debug?

How can feedback improve a generative AI solution?

Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Main topic: Agents

Scenario: The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.

Validation: You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

Lab 04: Objectives

Explain agent roles and use cases.

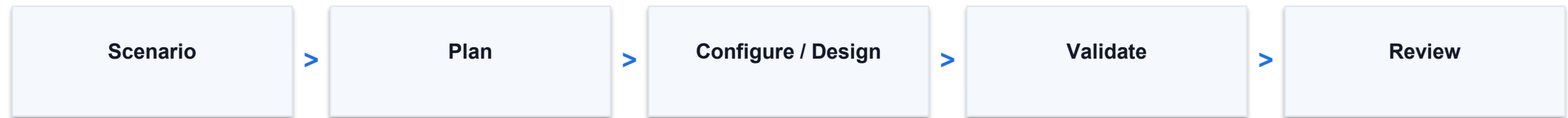
Plan Foundry Agent Service resources.

Design custom agent tools and workflows.

Review multi-agent orchestration concepts.

Lab 04: Lab Alignment Diagram

Lab-aligned design flow



Scenario

The support AI assistant needs to look up order status, search support documents, draft a reply, and escalate complex cases to a human agent.

Lab steps

- Define Agent Use Case
- Plan Agent Resources
- Design Tool Calls
- Design Orchestration
- Multi-Agent Review

Validation

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

Lab 04: Step Sequence

1. Define Agent Use Case
2. Plan Agent Resources
3. Design Tool Calls
4. Design Orchestration
5. Multi-Agent Review

Lab 04 Step 1: Define Agent Use Case

Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Write:

Agent goal:
User types:
Allowed tools:
Disallowed actions:
Human escalation trigger:
Success metric:

Record the required output in your lab notes and verify it before continuing.

Lab 04 Step 2: Plan Agent Resources

Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Document:

- Foundry project.
- Model deployment.
- Search tool.
- Function or API tools.
- Authentication.
- Logging and trace settings.

Record the required output in your lab notes and verify it before continuing.

Lab 04 Step 3: Design Tool Calls

Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Create a table:

Tool	Purpose	Input	Output
Search knowledge base	Retrieve support articles	Query	Relevant passages
Order lookup	Get order status	Order ID	Status summary

Record the required output in your lab notes and verify it before continuing.

Lab 04 Step 4: Design Orchestration

Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Draw:

User request -> agent plan -> tool call -> observation -> model reasoning -> response -> feedback

Record the required output in your lab notes and verify it before continuing.

Lab 04 Step 5: Multi-Agent Review

Agentic AI, Foundry Agent Service, Multi-Agent Concepts

Explain when separate agents might be useful for retrieval, billing, technical support, and escalation review.

Record the required output in your lab notes and verify it before continuing.

Lab 04: Validation and Checkpoints

You should have agent use case, resource plan, tool table, orchestration diagram, and multi-agent notes.

What is an AI agent?

Why do agents need tool permissions?

What is orchestration?

Why should autonomous actions be constrained?

Lab 05: Computer Vision, Custom Vision, Video Insights

Main topic: Vision

Scenario: The company wants to inspect product photos, detect damaged items, read text from images, and extract insights from product demo videos.

Validation: You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.

Lab 05: Objectives

Analyze images with Azure AI Vision concepts.

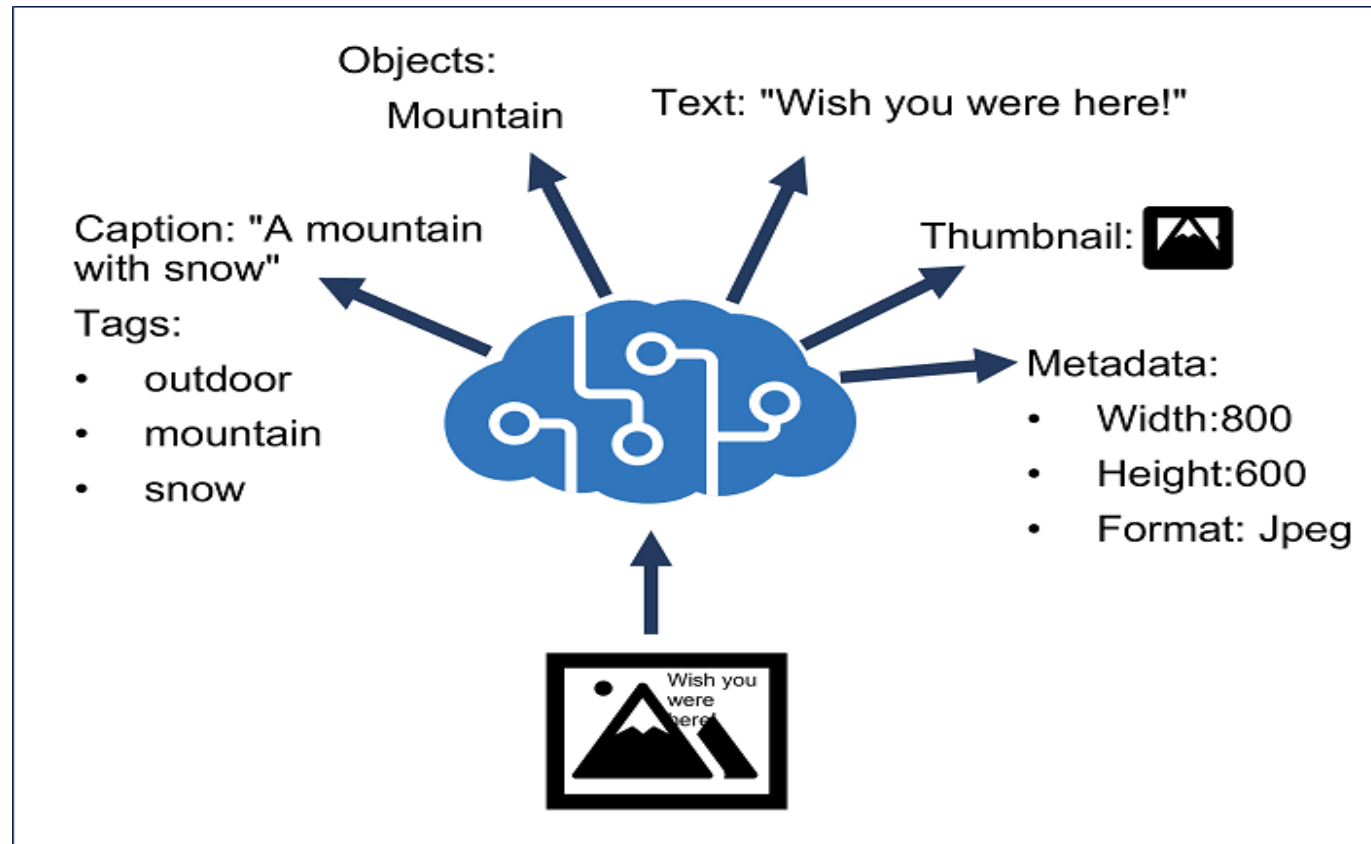
Select visual features for image processing requirements.

Compare image classification and object detection.

Plan custom vision training.

Explain video insight scenarios.

Lab 05: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
The company wants to inspect product photos, detect damaged items, read text from images, and extract insights...

Use it as a reference while completing the design steps.

Imported from source PPT slide 73.

FOLLOW IN ORDER

Lab 05: Step Sequence

1. Map Vision Requirements
2. Plan an Image Request
3. Custom Vision Plan
4. Video Insights Plan
5. Responsible AI Review

Lab 05 Step 1: Map Vision Requirements

Computer Vision, Custom Vision, Video Insights

Requirement	Vision Capability
Generate image tags	Image analysis
Detect product locations	Object detection
Read text from photos	OCR
Classify product condition	Image classification
Extract video topics	Video insights

Record the required output in your lab notes and verify it before continuing.

Lab 05 Step 2: Plan an Image Request

Computer Vision, Custom Vision, Video Insights

Document:

Image source
Visual features
Language
Expected response fields
Error handling
Privacy considerations

Record the required output in your lab notes and verify it before continuing.

Lab 05 Step 3: Custom Vision Plan

Computer Vision, Custom Vision, Video Insights

Choose:

Need	Model Type
One label for each image	Image classification
Locate multiple objects	Object detection

Document image labels, training images, evaluation metrics, publishing, and consumption.

Record the required output in your lab notes and verify it before continuing.

Lab 05 Step 4: Video Insights Plan

Computer Vision, Custom Vision, Video Insights

Explain how Video Indexer can extract:

- Transcript.
- Keywords.
- Faces or speakers where appropriate.
- Topics.
- Scenes.

Record the required output in your lab notes and verify it before continuing.

Lab 05 Step 5: Responsible AI Review

Computer Vision, Custom Vision, Video Insights

Document consent, retention, face analysis sensitivity, and human review needs.

Record the required output in your lab notes and verify it before continuing.

Lab 05: Validation and Checkpoints

You should have vision mapping, request plan, custom vision plan, video insight notes, and responsible AI review.

What is OCR?

How is object detection different from classification?

What does a custom vision model require?

Why is video analysis sensitive?

Lab 06: NLP, Language, Speech, Translation

Main topic: Language and speech

Scenario: The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generate spoken responses.

Validation: You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

Lab 06: Objectives

Analyze and translate text.

Extract key phrases, entities, PII, language, and sentiment.

Process speech with speech-to-text and text-to-speech.

Plan translation and SSML use.

Lab 06: Imported Reference Diagram

Feature	Image
Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/speech-to-text/about
Custom Speech to text	mcr.microsoft.com/product/azure-cognitive-services/speechservices/custom-speech-to-text/about
Neural Text to speech	mcr.microsoft.com/product/azure-cognitive-services/speechservices/neural-text-to-speech/about
Speech language detection	mcr.microsoft.com/product/azure-cognitive-services/speechservices/language-detection/about

How to use it

Connect the diagram to the lab scenario:
The support platform must analyze customer messages, detect PII, translate text, transcribe calls, and generat...

Use it as a reference while completing the design steps.

Imported from source PPT slide 61.

Lab 06: Step Sequence

1. Map Language Tasks
2. Design Text Analysis Pipeline
3. Design Speech Pipeline
4. SSML Review
5. Translation Review

Lab 06 Step 1: Map Language Tasks

NLP, Language, Speech, Translation

Scenario	Capability
Identify customer sentiment	Sentiment analysis
Extract product names	Entity recognition
Identify important topics	Key phrase extraction
Detect personal data	PII detection
Detect message language	Language detection
Translate message	Translator

Record the required output in your lab notes and verify it before continuing.

Lab 06 Step 2: Design Text Analysis Pipeline

NLP, Language, Speech, Translation

Input text -> language detection -> PII detection -> sentiment -> entities -> key phrases -> route case

Record the required output in your lab notes and verify it before continuing.

Lab 06 Step 3: Design Speech Pipeline

NLP, Language, Speech, Translation

Audio input -> speech to text -> intent or keyword recognition -> response -> text to speech

Record the required output in your lab notes and verify it before continuing.

Lab 06 Step 4: SSML Review

NLP, Language, Speech, Translation

Explain how SSML can control voice, pitch, rate, pauses, and pronunciation.

Record the required output in your lab notes and verify it before continuing.

Lab 06 Step 5: Translation Review

NLP, Language, Speech, Translation

Document:

- Text translation.
- Document translation.
- Speech-to-text translation.
- Speech-to-speech translation.
- Human review for critical messages.

Record the required output in your lab notes and verify it before continuing.

Lab 06: Validation and Checkpoints

You should have language task mapping, text pipeline, speech pipeline, SSML notes, and translation notes.

What is entity recognition?

Why detect PII before storing text?

What is SSML?

When should translation outputs be reviewed?

AI-102 INTEGRATION AND REVIEW

Day 2

Custom language, search, document intelligence, capstone and assessment.

Lab 07: Custom Language Models and Question Answering

Main topic: Custom language

Scenario: The support assistant must recognize customer intents and answer frequently asked questions from approved sources.

Validation: You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.

Lab 07: Objectives

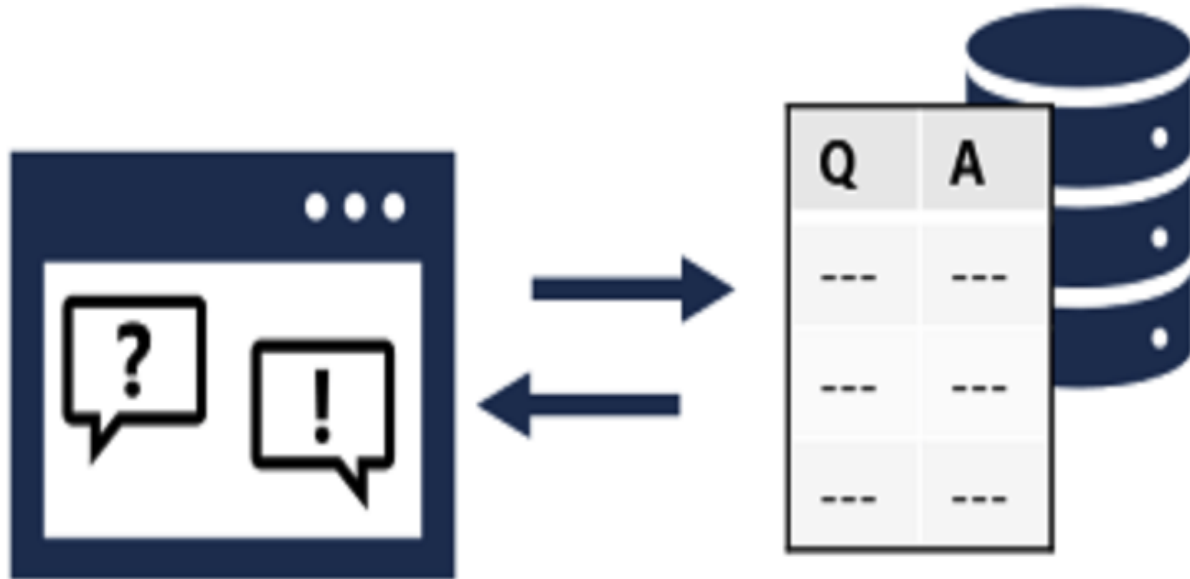
Design custom language understanding.

Define intents, entities, and utterances.

Plan custom question answering.

Create a multilingual question answering strategy.

Lab 07: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
The support assistant must recognize customer intents and answer frequently asked questions from approved sour...

Use it as a reference while completing the design steps.

Imported from source PPT slide 149.

Lab 07: Step Sequence

1. Define Intents and Entities
2. Plan Training Data
3. Plan Custom Question Answering
4. Multilingual Strategy
5. Backup and Recovery

Lab 07 Step 1: Define Intents and Entities

Custom Language Models and Question Answering

Create a table:

Intent	Example Utterance	Entities
CheckOrderStatus	Where is order 12345?	order number
RequestRefund	I want a refund for my laptop	product

Record the required output in your lab notes and verify it before continuing.

Lab 07 Step 2: Plan Training Data

Custom Language Models and Question Answering

Document:

- Number of example utterances.
- Balanced examples across intents.
- Entity labeling rules.
- Test set.
- Ambiguous utterances.

Record the required output in your lab notes and verify it before continuing.

Lab 07 Step 3: Plan Custom Question Answering

Custom Language Models and Question Answering

Record:

Knowledge sources
Q&A pairs
Alternate phrasing
Chit-chat
Multi-turn prompts
Publishing target
Fallback answer

Record the required output in your lab notes and verify it before continuing.

Lab 07 Step 4: Multilingual Strategy

Custom Language Models and Question Answering

Explain whether to translate sources, create separate projects, or translate user input before answering.

Record the required output in your lab notes and verify it before continuing.

Lab 07 Step 5: Backup and Recovery

Custom Language Models and Question Answering

Document export, versioning, backup, and recovery of language projects.

Record the required output in your lab notes and verify it before continuing.

Lab 07: Validation and Checkpoints

You should have intent/entity table, training plan, question answering plan, multilingual strategy, and backup notes.

What is an intent?

What is an entity?

Why add alternate phrasing?

Why should language projects be backed up?

Lab 08: Azure AI Search, Knowledge Mining, Vector Search

Main topic: Search and mining

Scenario: The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vector embeddings for RAG.

Validation: You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

Lab 08: Objectives

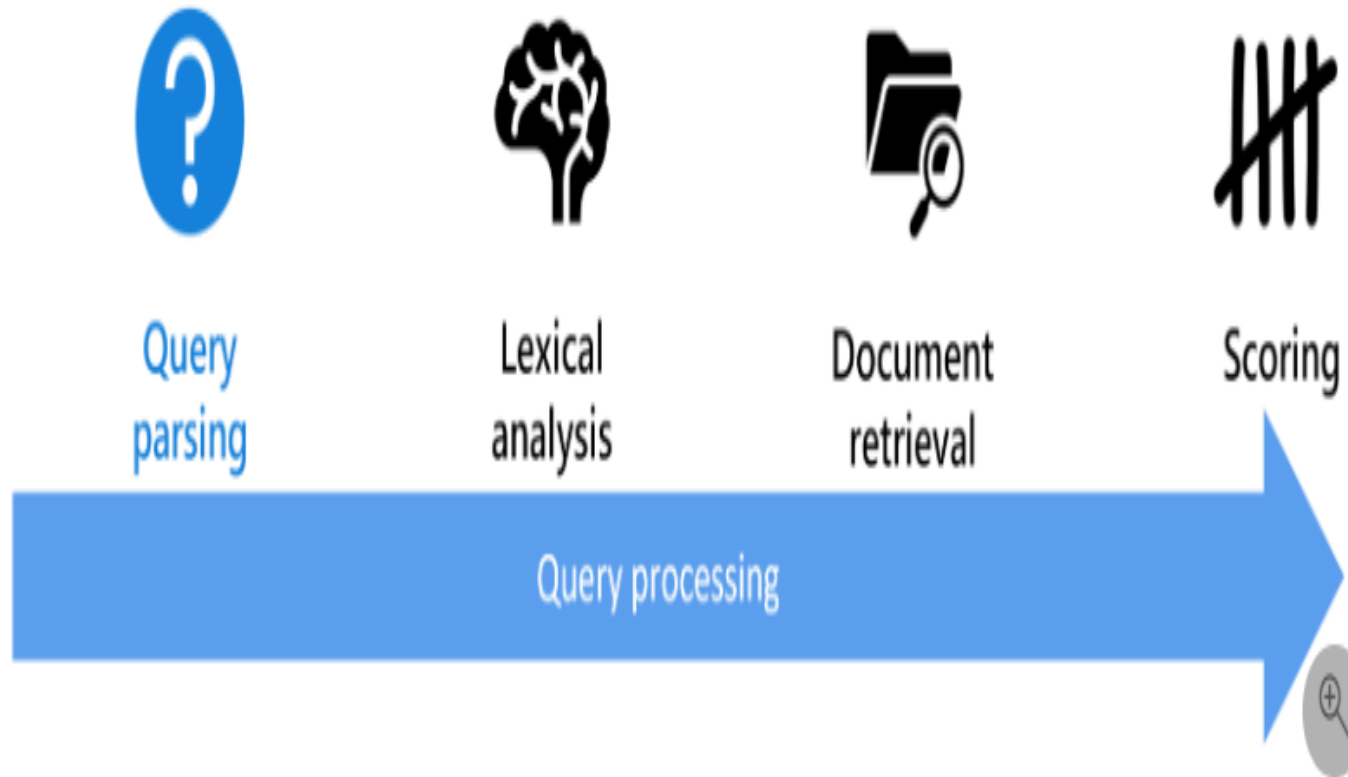
Design an Azure AI Search solution.

Define indexes, data sources, indexers, and skillsets.

Explain semantic and vector search.

Plan knowledge store projections.

Lab 08: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
The company wants searchable support documents, enriched with extracted text, entities, key phrases, and vecto...

Use it as a reference while completing the design steps.

Imported from source PPT slide 380.

Lab 08: Step Sequence

1. Design an Index
2. Plan Data Sources and Indexers
3. Define a Skillset
4. Query Design
5. Knowledge Store Projections

Lab 08 Step 1: Design an Index

Azure AI Search, Knowledge Mining, Vector Search

Define fields:

```
id  
title  
content  
category  
source_url  
last_updated  
security_label  
embedding
```

Record the required output in your lab notes and verify it before continuing.

Lab 08 Step 2: Plan Data Sources and Indexers

Azure AI Search, Knowledge Mining, Vector Search

Document:

- Source storage.
- Supported document formats.
- Indexer schedule.
- Change detection.
- Error handling.

Record the required output in your lab notes and verify it before continuing.

Lab 08 Step 3: Define a Skillset

Azure AI Search, Knowledge Mining, Vector Search

Include:

- OCR.
- Language detection.
- Entity extraction.
- Key phrase extraction.
- Custom skill.
- Embedding generation.

Record the required output in your lab notes and verify it before continuing.

Lab 08 Step 4: Query Design

Azure AI Search, Knowledge Mining, Vector Search

Write examples for:

Keyword search
Filter by category
Sort by last updated
Wildcard query
Semantic ranking
Vector search
Hybrid search

Record the required output in your lab notes and verify it before continuing.

Lab 08 Step 5: Knowledge Store Projections

Azure AI Search, Knowledge Mining, Vector Search

Explain file, object, and table projections and when enriched output should be stored.

Record the required output in your lab notes and verify it before continuing.

Lab 08: Validation and Checkpoints

You should have index design, data source/indexer plan, skillset plan, query examples, and knowledge store notes.

What is an indexer?

What is a skillset?

What is vector search?

Why use semantic ranking?

Lab 09: Document Intelligence and Content Understanding

Main topic: Document AI

Scenario: The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to extract structured information from multiple content types.

Validation: You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

Lab 09: Objectives

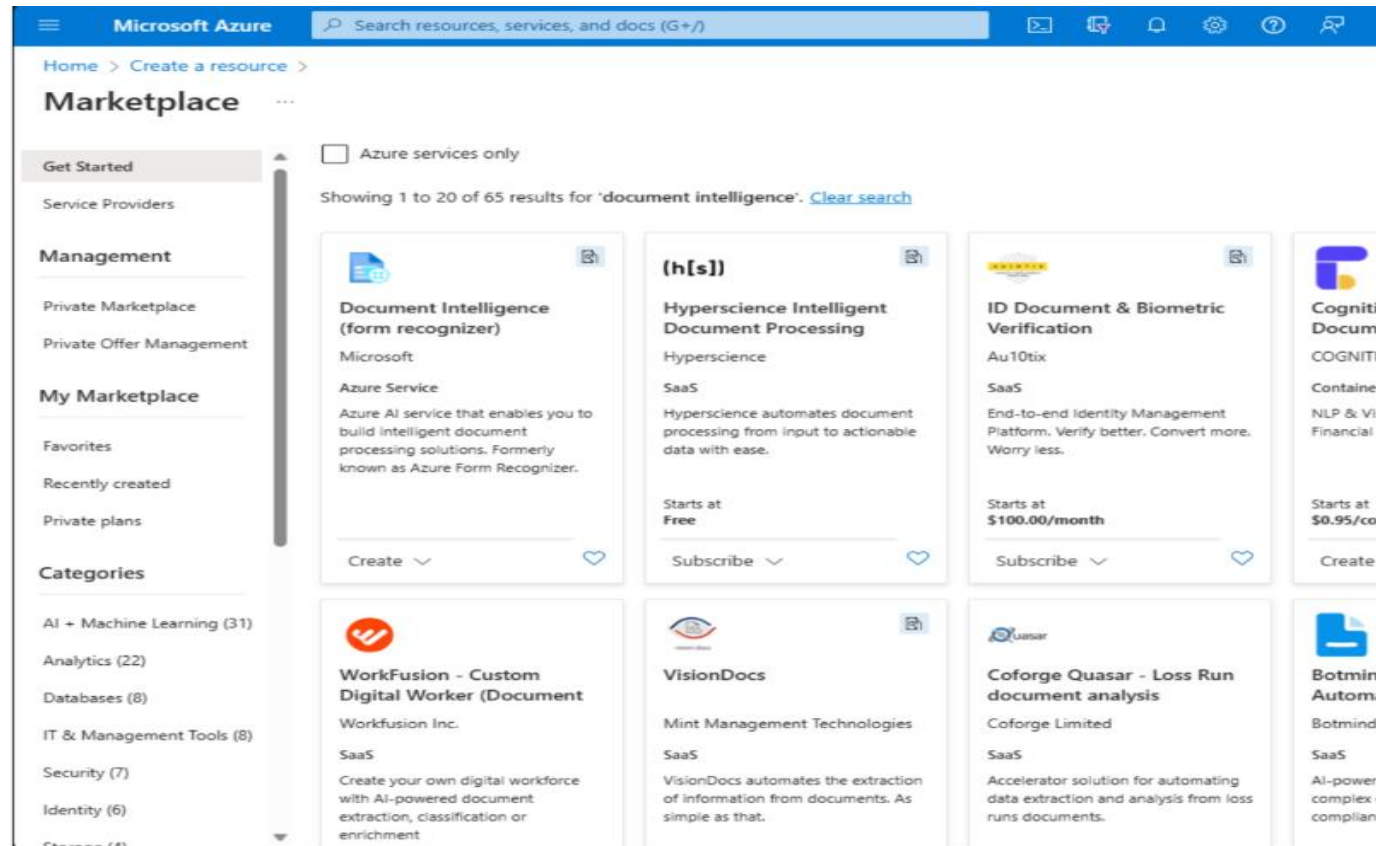
Use prebuilt document extraction concepts.

Plan custom Document Intelligence models.

Explain composed models.

Design Content Understanding extraction workflows.

Lab 09: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
The company processes invoices, contracts, support screenshots, audio notes, and product videos. It wants to e...

Use it as a reference while completing the design steps.

Imported from source PPT slide 532.

Lab 09: Step Sequence

1. Map Document Scenarios
2. Custom Document Model Plan
3. Content Understanding Plan
4. Human Review Rules
5. Integration Flow

Lab 09 Step 1: Map Document Scenarios

Document Intelligence and Content Understanding

Scenario	Capability
Extract invoice fields	Prebuilt invoice model
Extract custom form fields	Custom document model
Route different form types	Composed model
Extract table data	Layout and table extraction
Extract text from images	OCR pipeline

Record the required output in your lab notes and verify it before continuing.

Lab 09 Step 2: Custom Document Model Plan

Document Intelligence and Content Understanding

Document:

Document types
Sample count
Fields to extract
Labels
Training split
Evaluation metric
Publish plan

Record the required output in your lab notes and verify it before continuing.

Lab 09 Step 3: Content Understanding Plan

Document Intelligence and Content Understanding

Explain extraction from:

- Documents.
- Images.
- Video.
- Audio.

Include summarization, classification, attributes, entities, tables, and images.

Record the required output in your lab notes and verify it before continuing.

Lab 09 Step 4: Human Review Rules

Document Intelligence and Content Understanding

Define confidence thresholds and when extracted data requires review.

Record the required output in your lab notes and verify it before continuing.

Lab 09 Step 5: Integration Flow

Document Intelligence and Content Understanding

Draw:

Upload content -> extract text/entities/tables -> validate -> store structured data -> search/report

Record the required output in your lab notes and verify it before continuing.

Lab 09: Validation and Checkpoints

You should have scenario mapping, custom model plan, content understanding plan, review rules, and integration flow.

When should you use a prebuilt model?

What is a composed model?

Why are confidence scores useful?

What content types can Content Understanding process?

Lab 10: AI-102 Capstone and Course Review

Main topic: Integrated review

Scenario: You must design a production-ready customer support AI platform that supports grounded chat, document search, sentiment analysis, speech transcription, image OCR, document extraction, and human escalation.

Validation: You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

Lab 10: Objectives

Design an end-to-end Azure AI solution.

Map requirements to Microsoft Foundry and Azure AI services.

Include security, monitoring, responsible AI, and cost controls.

Build a personal review plan.

Lab 10: Imported Reference Diagram



How to use it

Connect the diagram to the lab scenario:
You must design a production-ready customer support AI platform that supports grounded chat, document search, ...

Use it as a reference while completing the design steps.

Imported from source PPT slide 716.

Lab 10: Step Sequence

1. Build a Service Map
2. Draw the Architecture
3. Add Governance Controls
4. Rate Skill Confidence
5. Clean Up Resources
6. Create a 7-Day Review Plan

Lab 10 Step 1: Build a Service Map

AI-102 Capstone and Course Review

Requirement	Service
Grounded assistant	Foundry, Azure OpenAI, prompt flow, RAG
Agent workflow	Foundry Agent Service
Document search	Azure AI Search
Sentiment and PII	Azure AI Language
Speech transcription	Azure AI Speech
Image OCR	Azure AI Vision
Invoice extraction	Document Intelligence
Multi-content extraction	Content Understanding
Safety controls	Content Safety and responsible AI governance

Record the required output in your lab notes and verify it before continuing.

Lab 10 Step 2: Draw the Architecture

AI-102 Capstone and Course Review

Include:

User application
API layer
Foundry project
Model deployment
AI Search index
Storage
Language/Speech/Vision/Document services
Monitoring
Human review queue

Record the required output in your lab notes and verify it before continuing.

Lab 10 Step 3: Add Governance Controls

AI-102 Capstone and Course Review

Document:

- RBAC.
- Managed identity.
- Key protection.
- Prompt filters.
- Content safety.
- Logging and monitoring.
- Budget alerts.
- Data retention.

Human escalation

Record the required output in your lab notes and verify it before continuing.

Lab 10 Step 4: Rate Skill Confidence

AI-102 Capstone and Course Review

Skill Area	Confidence 1-5	Study Action
Plan and manage Azure AI solution		
Implement generative AI solutions		
Implement agentic solution		
Implement computer vision solutions		
Implement NLP solutions		
Implement knowledge mining and information extraction		

Record the required output in your lab notes and verify it before continuing.

Lab 10 Step 5: Clean Up Resources

AI-102 Capstone and Course Review

If you created resources:

```
az group delete --name rg-ai102-lab --yes --no-wait
```

Confirm with your instructor before deleting shared resources.

Record the required output in your lab notes and verify it before continuing.

Lab 10 Step 6: Create a 7-Day Review Plan

AI-102 Capstone and Course Review

- Day 1: Planning, security, monitoring, responsible AI.
- Day 2: Generative AI, prompt flow, RAG.
- Day 3: Agents and orchestration.
- Day 4: Vision and video.
- Day 5: NLP, speech, translation, question answering.
- Day 6: AI Search, Document Intelligence, Content Understanding.
- Day 7: Capstone review and mistakes log.

Record the required output in your lab notes and verify it before continuing.

Lab 10: Validation and Checkpoints

You should have a service map, architecture diagram, governance checklist, confidence matrix, cleanup plan, and review plan.

Which service should power grounded document search?

When should a human review AI output?

What is the difference between RAG and fine-tuning?

Which AI workload do you need to review most?

COURSE CLOSING

Assessment and Wrap-up

Take the Online AI-102 Practice Exam

- Go to the Tertiary Exams portal and open the matching AI-102 Practice Exam.
- Use the practice set to test Azure AI planning, Azure OpenAI, search, vision, language and document intelligence.
- Review mode lets you check the answer and revisit the concept after each question.
- Complete the practice exam before assessment, then bring difficult items to the review session.

Practice exam link:

<https://exams.tertiaryinfotech.com>

The screenshot shows the Tertiary Exams portal interface. At the top, there's a navigation bar with 'Tertiary Exams', 'Practice Exams', 'About Us', 'Sign In', and a 'Get started' button. Below the navigation bar, the main heading is 'Microsoft Certified Azure AI Engineer Associate AI-102 Training Practice Exam'. Underneath, it says 'Practice questions for Azure AI engineering, responsible AI, retrieval, agents and multimodal services.' There are three tabs: 'MODE' (Practice), 'FOCUS' (AI-102), and 'ACCESS' (Portal link). To the right, there's a 'PRACTICE DETAIL' box with 'Exam code AI-102', 'Use Review', 'Domains Azure AI', and 'Format Online'. Below the tabs, there are two main sections: 'Practice mode' (Check answers while revising each domain) and 'Review before WA' (Use missed questions to guide revision). On the far right, there's a 'Learner tip' box (Retake missed items after revision).

Assessment Flow Reminder



The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Certification and TRAQOM Survey

Complete the certification and TRAQOM survey to receive your certificate.
Remember AM, PM and assessment digital attendance.
Contact support if you need help after class.

Support

Email: enquiry@tertiaryinfotech.com

Tel: +65 6100 0613

Website: www.tertiarycourses.com.sg

CONGRATULATIONS

Thank You

You have designed an end-to-end Azure AI engineering solution.